

Product lifecycle oriented digitization agile process preparation system

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Abstract

After analyzing the features of process preparation, a product lifecycle oriented digitization agile process preparation system (DAPP) is presented. This paper develops a general approach and a system tool for digitization agile process preparation (DAPP). First, according to modern manufacturing concept and the bottleneck problems of computer integrated manufacturing system (CIMS), a concept of DAPP is proposed. Secondly, after analysis of the characteristics of the process preparation system (DAPPS), the needs for DAPPS are explored in depth. Thirdly a framework is proposed to support DAPPS by providing computer-aided and database environments for DAPPS, advisory tools for product structure manufacturability evaluation, rapid process planning, tooling design, process consumption determination as well as field process preparation, and knowledge bases incorporated in the advisory tools. Fourthly, according to the framework, a design support strategy for DAPPS is investigated to guide the coordination of preparation activities. Finally, based on the proposed approach and strategy, the system prototype has been developed employing the knowledge-based approach. The results of this research will facilitate the rationalization of process preparation and thus shorten the preparation time.

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Keywords: Process preparation; Product lifecycle; Process planning; Knowledge reuse and recombination

1. Introduction

Process preparation is an important interface between computer-aided design (CAD) and computer-aided manufacturing (CAM) in the computer integrated manufacturing environment. Process preparation involves a wide spectrum of activities that include product structure manufacturability evaluation, technological process design, tooling design, production process preparation and production process organization, etc. [1], the core of which is to manage and take full advantage of enterprise information resources so as to enhance the rapid response ability and economic utility of enterprises.

Present process preparation studies mostly focus on process planning, especially on computer-aided process planning (CAPP). World-famous enterprises, such as Boeing, Lockheed, Airbus and so on, have done great research on management and optimization of technology and process, developed and integrated a great quantity of CAPP engineering application software and manufacturing data management software, established many engineering data bases, material bases,

design and manufacturing feature data bases, typical process bases, and preliminarily realized the information sharing. Altung and Zhang presented a survey of about 206 references and an extensive list of CAPP systems developed until 1988 [2]. Most of traditional systems can only provide one process plan per part design [3] and thus the flexibility and responsiveness will descend [4]. The research issues in the area of CAPP were brought out in a survey article by Cay and Chasspis [5]. With the development of computer technology, digitization technology and intellectual technology, the development trends of CAPP are: (1) intellectual CAPP. Some systems have been researched including expert system-based CAPP [6], artificial neural net-based CAPP [7], case-based reasoning CAPP [8] and other intellectual technology-based system which play an active role in extending information processing capacity and functions of CAPP. (2) CAPP under variant manufacturing patterns including integration CAPP, concurrent CAPP, distributed and cooperative CAPP and so on.

Besides process planning, information modeling and sharing are also worth studying. In order to integrate the whole product life cycle information and realize data sharing which are the bottleneck problems of computer integrated manufacturing system, many researchers have carried out a great deal of investigation. Bhandarkar and Nagi developed a feature

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information extraction system based on B-rep and a series of feature identification methods with geometric topology input and STEP files with form feature information represented by AP224 output [9]. Liu et al. presented an automatic feature extraction system (AFES) that translates data in a CAD file into manufacturing features by concave edge test, convex feature extraction and minimal enclosing box feature extraction [10]. Subrahmanyam explored heuristic-based method for volume

decomposition and combination to obtain manufacturing features by [11]. In terms of information modeling, many models have been presented aiming at variant data exchange standards, aided modeling techniques, model types, modeling environments and so on.

Despite of these above achievements, the problems of process preparation may be summarized as follows [2,4,5, 8–10]:

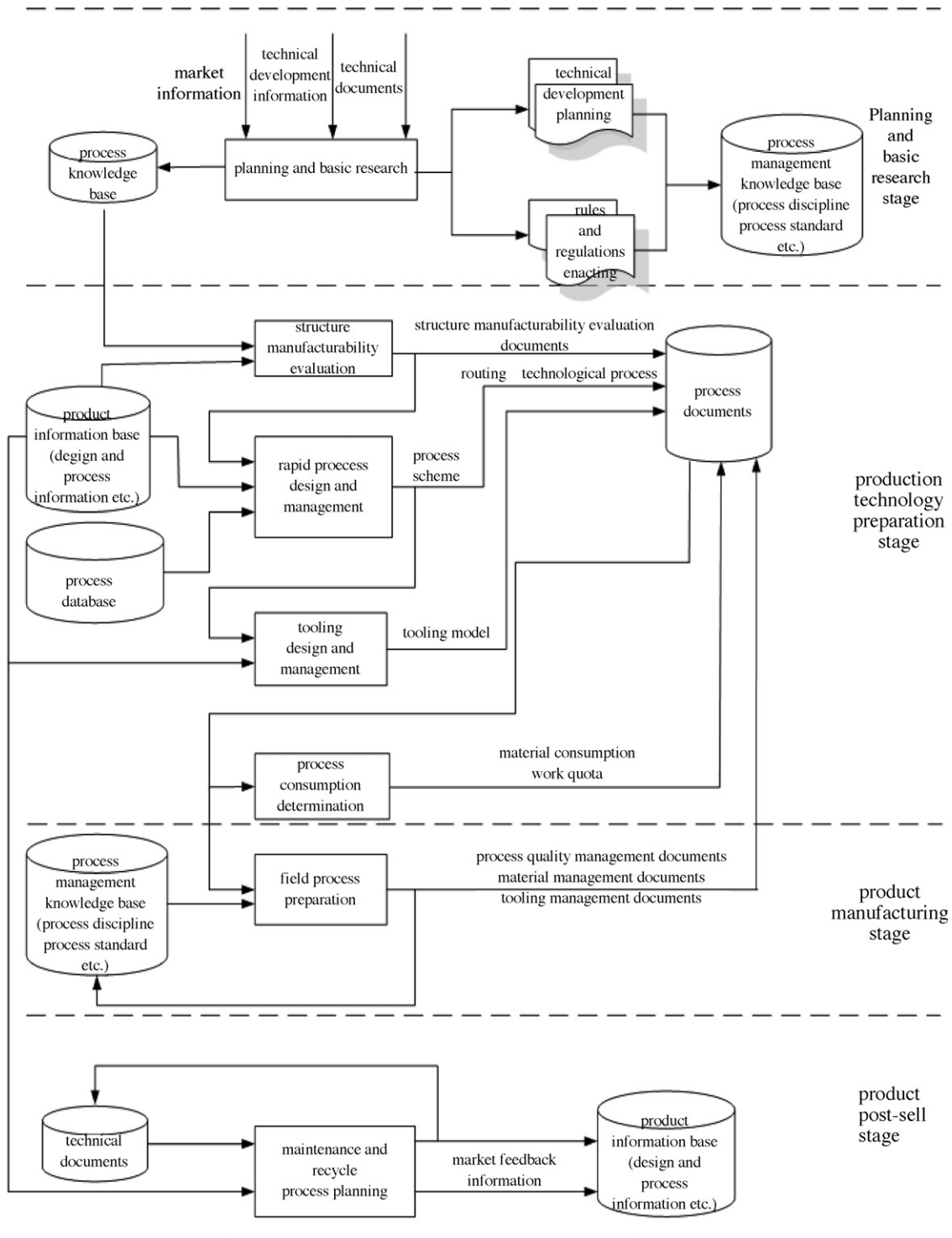
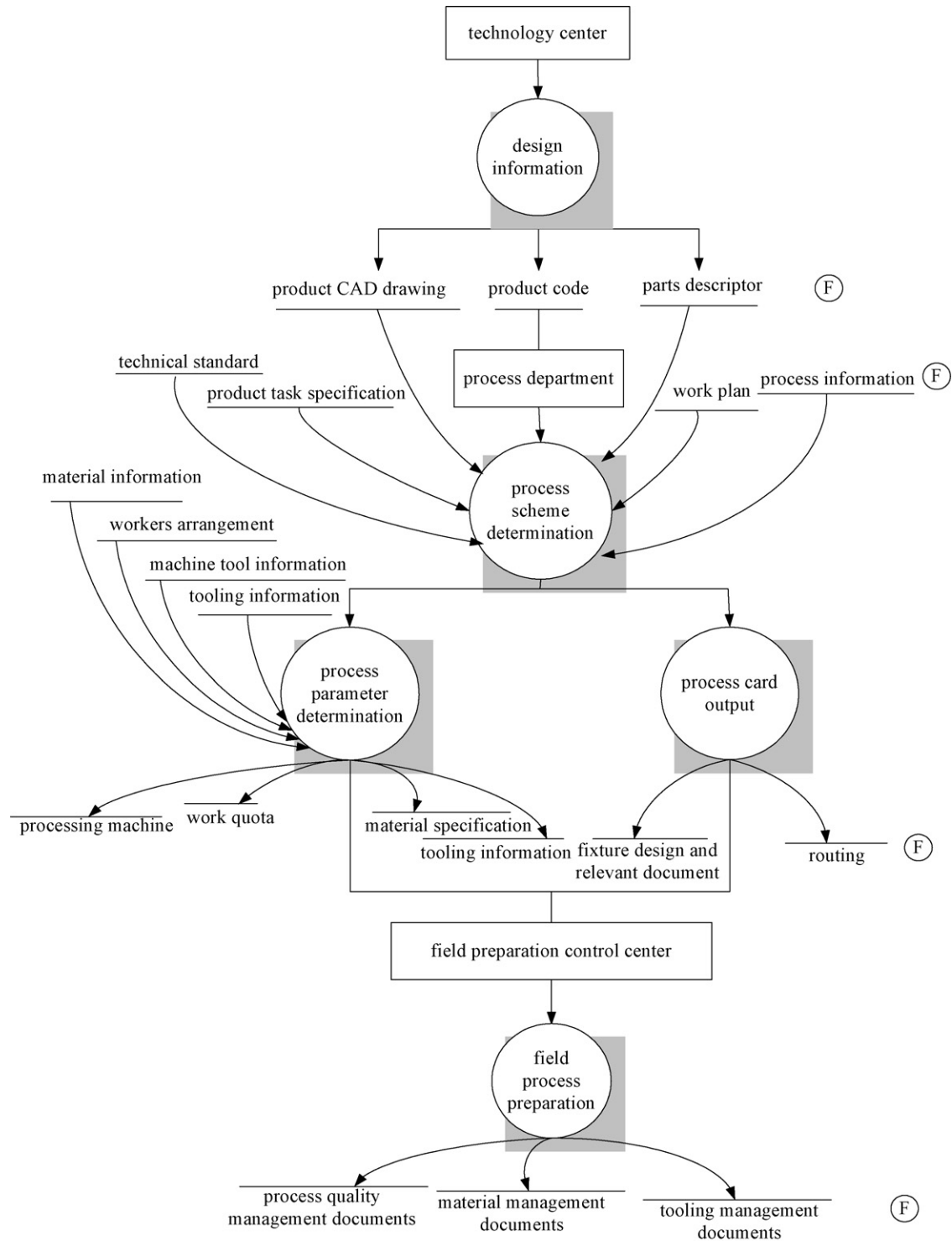


Fig. 1. Architectural structure diagram of DAPPS.

- (a) Absence of whole product life cycle oriented information integration and sharing and failure to make information flexible.
- (b) Insufficient for universality.
- (c) Information with big draining, poor smoothness and deficient integrity.

To resolve the above problems, for process preparation activity it is necessary to develop a system to guide the coordination of activities and support the decision-making at each preparation stage of whole product life cycle. The present work develops a general approach and a system tool for digitization agile process preparation (DAPP) to facilitate



Ⓡ: The output may be feedback to relevant departments according to requirements.

Fig. 2. Data flow diagram of DAPPS.

the rationalization of process preparation and thus shorten the preparation time.

2. Definition of DAPP and analysis of the characteristics of DAPPS

Digitization agile process preparation means to digitize and computerize the process preparation work during the whole product lifecycle from product design and product manufacturing to production organization on the basis of the process standard, and thus generate institutive knowledge and achieve the aim of shortening the preparation time, optimizing the process planning and reducing the cost of product development.

Different preparers involve the constraints from different viewpoints in the basic digitization agile process preparation activities carried out. The activities interaction makes the preparation a complex task to realize all requirements successfully. The interaction relationship among digitization agile process preparation is described in Fig. 1.

Depending on the stage of the preparation process, the interaction level between activities may be different. The digitization agile process preparation flow must be analyzed and modeled in a structured way. The data flow diagram (DFD) of DAPPS is always given by the structured analysis (SA) method because SA is useful for requirement analysis. SA is a popular requirement analysis technique. This method is data flow oriented from top to down and stepwise hierarchical decomposition. SA obtains high-level data flow from requirement, then defines and stores the data flow and data to the element level by analysis of I/O of DFD, and finally determines the fundamental element of the system [12]. The data flow of DAPPS using the structured analysis method is illustrated in Fig. 2.

The characteristics of DAPPS are as follows:

- Product lifecycle oriented process preparation.* In DAPPS, conventional process preparations are extended and divided into four stages from the viewpoint of process work during product lifecycle: planning and basic research, production technology preparation, product manufacturing and product post-sell stage.
- Product lifecycle oriented product digitization information modeling.* Whole process preparation work is incorporated by product information model.
- Agility.* DAPPS can realize agile digitization process preparation in two ways. One is the optimal production procedure organization and integration. Concurrent process preparation and data sharing between the upper stream and the down stream of process preparation are effectively carried out by reasonable planning and organization of the process preparation work during life cycle and collaborative process preparation are also carried out based on network technology. The other is the implementation technology. The functions of DAPPS can be realized by reusing and recombining enterprises' outside resources (i.e., market information, technology development home and abroad, etc.) and inside resources (i.e., product information, original technical documents, process knowledge, equipment, human resource, etc.). Also reconfigurable and scalable functional module should be included.

3. Design support system for digitization agile process preparation

3.1. System framework

Owing to the complicated interactions among the activities of process preparation, the bottleneck problems of CIMS and the rapid development of digitization techniques and computer-aided technique, there is a need to develop a systematic design

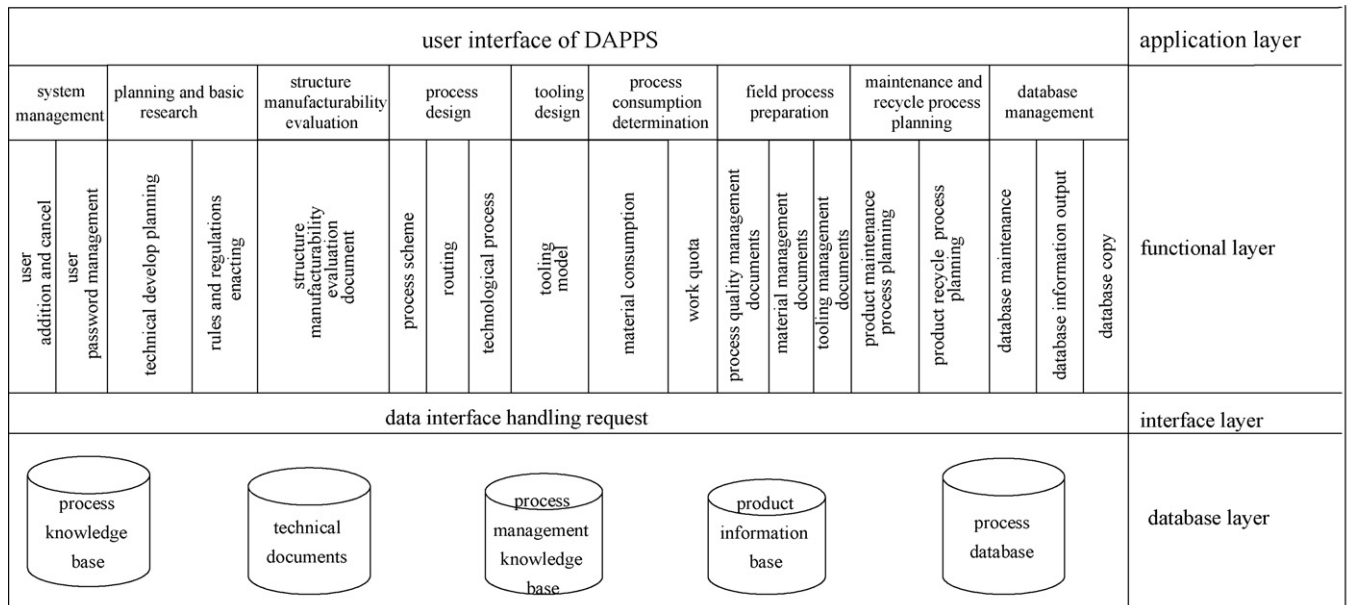


Fig. 3. System framework of DAPPS.

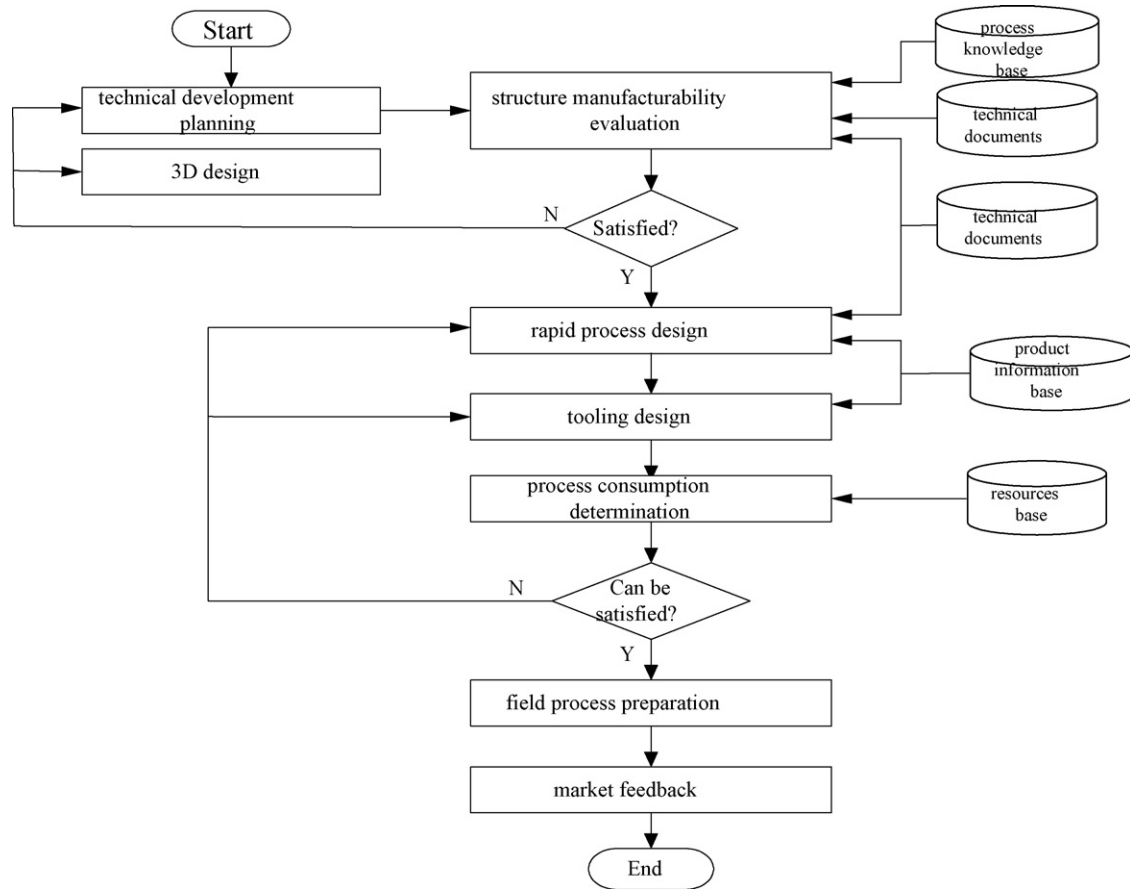


Fig. 4. Design support strategy for DAPPS.

support framework to guide the coordination of activities and to support the decision-making at each preparation stage. According to the functional requirements, a framework for DAPPS is proposed, as shown in Fig. 3. This system framework divides the process preparation into four stages from the viewpoint of process work during the whole product lifecycle: planning and basic research, production technology preparation, product manufacturing and product post-sell stage, which captures the key issues of the preparation process and is tailored for use by a preparer who is familiar with some function aspects but less experienced with others. The framework mainly covers:

- User management.* By setting user roles of different level and corresponding power, users of different geographic position or different function can be managed effectively.
- Product structure manufacturability evaluation.
- Computer-aided process planning.
- Field process preparation.
- Maintenance and recycle process planning.
- Knowledge bases incorporated in the advisory tools.

Two of the major elements in DAPPS are the structure manufacturability evaluation and rapid process planning at the early and mid process preparation stage. The knowledge base can easily incorporate the structure manufacturability evaluation and typical process, which will enable the preparer to

improve and evaluate alternative schemes. The process preparation can be accelerated by the advisory tools, and the knowledge base equips the advisory tools to handle various preparation problems.

3.2. Design support strategy for digitization agile process preparation system

According to the system framework, the design support strategy shown in Fig. 4 is proposed to implement digitization agile process preparation, which aims to:

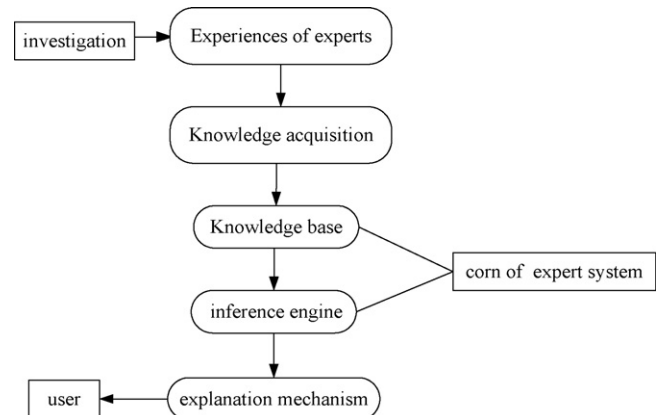


Fig. 5. Basic structure of expert system.

Table 1
Comparison and analysis of popular reasoning methods

Type	Method	Feature
Data driven	CBR (case-based reasoning)	Advantages: self-learning capability and good expressive power; disadvantages: need for great numbers of cases
	Neural net	Advantages: good numerical processing and self-learning, self-organizing, fault tolerance capability; disadvantages: need for great numbers of samples training
Knowledge driven	Reasoning by analogy	Advantages: easier to realize creativity; disadvantages: abstract knowledge and hard to adjudge similarity
	Reasoning by rules	Advantages: easier to realize; disadvantages: hard knowledge acquisition
	Qualitative reasoning	Advantages: easier to realize; disadvantages: hard knowledge acquisition
	Agent-based reasoning	Advantages: well-distributed processing; disadvantages: complex knowledge operation
	Evolutionary reasoning	Advantages: concurrent and good flexibility; disadvantages: long time to find the solution
	Feature-based reasoning	Advantages: well-distributed processing; disadvantages: poor intelligence

- (a) Trigger structure manufacturability to provide structure manufacturability feedback in response to functional design changes.
- (b) Rapidly design technological process by means of process data digging and reasoning base on reusing and recombining process knowledge.
- (c) Enable the designer to be aware of the feedback information.

The following summarizes the key issues of the design support strategy:

- (a) Product structure manufacturability evaluation. Calculate every index of structure manufacturability, evaluate the structure manufacturability, and then choose optimal or near-optimal structure proposal. In addition the output will be fed back to design department when necessary.
- (b) Rapid process planning. Searching for typical process based on obtained product information and technical documents, process scheme, process route and technological process are rapidly compiled by means of process data digging and reasoning based on reusing and recombining process knowledge.
- (c) Field process preparation. According to relevant documents, materials delivery, production layout design, process

quality control design, etc., of production field process preparation are carried out. Likewise the output can be used to guide following field process preparation as know-how.

- (d) Maintenance and recycle process planning. After product sale, according to the feedback usage information, maintenance and recycle process are planned utilizing existed techniques and documents, the result of which can be used to guide the process of existed product manufacturing and fresh product development.

3.3. Rapid process planning

3.3.1. Expert reasoning system

Expert reasoning system is an intelligent software system for problem solving which uses specialist reasoning and control strategy to solve problems with the level of expert group based on computerized and digitized knowledge and experiences of experts. An entire expert system consists of knowledge base, inference engine, knowledge acquisition and explanation interface, as shown in Fig. 5. To be simple, knowledge + algorithm = expert system. The key to create knowledge base is to represent knowledge, and the main issue of construct inference engine is to determine imprecise reasoning. The comparison and analysis of popular reasoning methods is shown in Table 1.

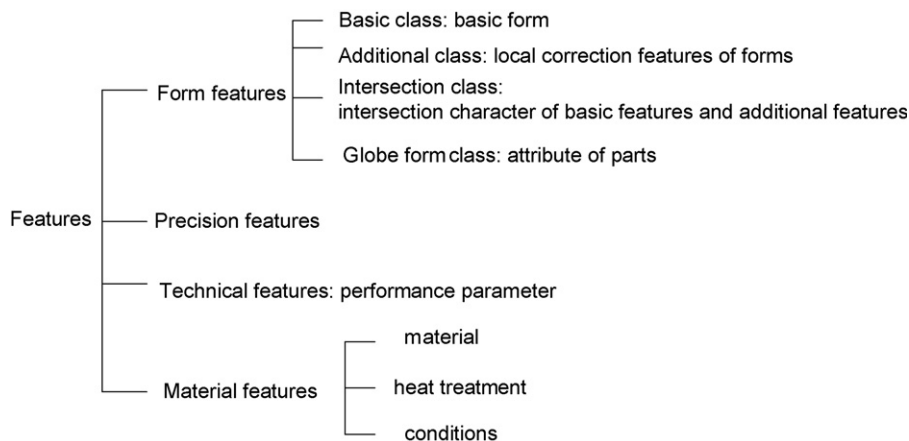


Fig. 6. Classification of features.

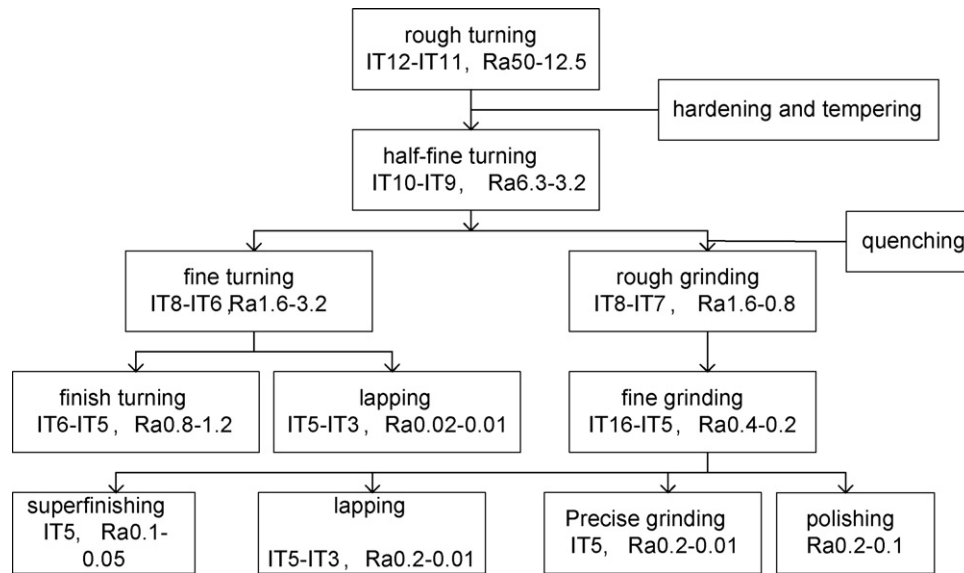


Fig. 7. typical process scheme of cylinder.

3.3.2. Feature-based reasoning

Feature is the definition of components' basic geometry entity for design or manufacturing, which is composed of plane, loop, edge, point, central line, central point and so forth. By changing size of finite features, infinite shape with certain engineering semantics can be constructed. Features are affiliation with processing tightly. Owing to improvement of

process technology, some new geometry entities can be defined as features, while process technology can be enhanced because of new feature to meet the requirement of structural function. Meanwhile features are extensible, however, a series of features are stable relatively during certain period, which are called standard features. The classification of features is described in Fig. 6.

All of the above features influence the determination of process technology among which form features do maximally and precision feature in the second place. After investigation on certain enterprise, popular form features are obtained including cylinder, hole, plane, external thread, internal thread, gear, cam, keyway and so on. Typical features have corresponding typical process scheme. For example, process plan of cylinder is as shown in Fig. 7.

3.3.3. Realization of systematic inference engine

Reasoning of systematic inference engine is mainly based on features of parts. First, extract parts features in demand automatically, and then output basic process scheme utilizing typical process scheme incorporated in knowledge base. In this

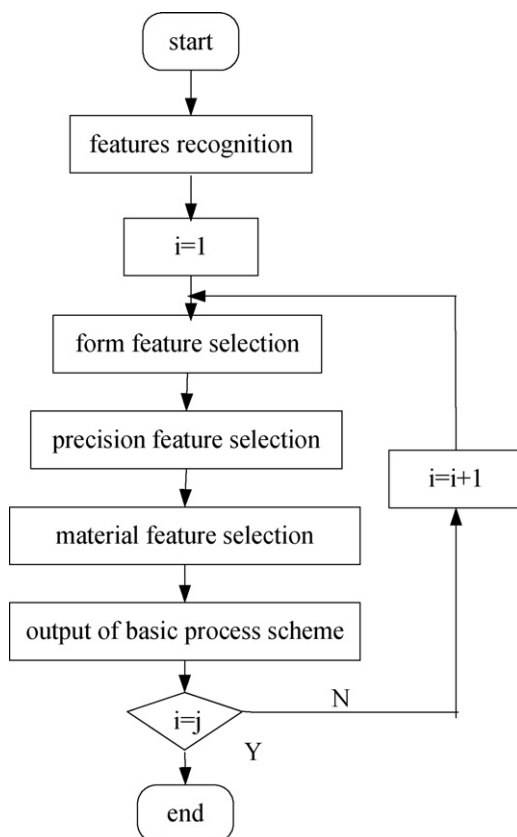


Fig. 8. Flow diagram of reasoning.

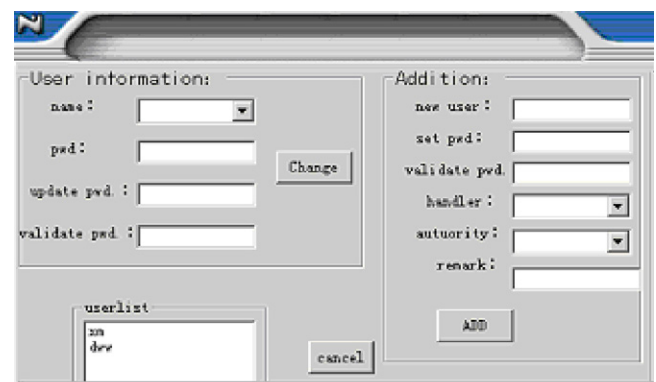


Fig. 9. System management interface.

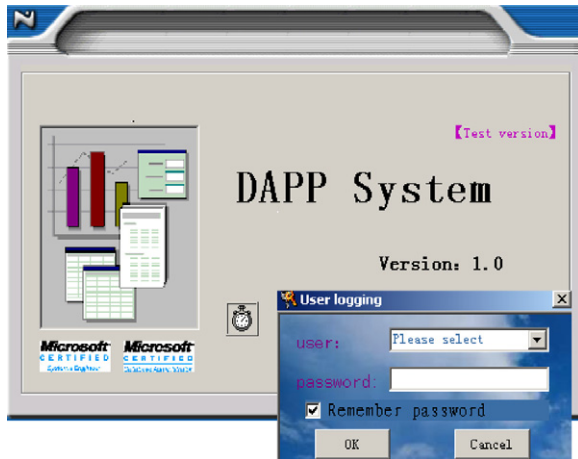


Fig. 10. System initial interface and user logging.

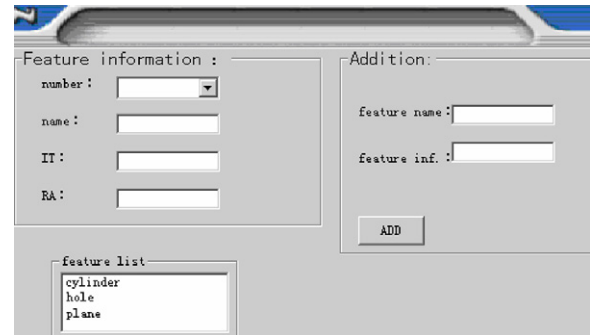


Fig. 12. Feature extracting interface.

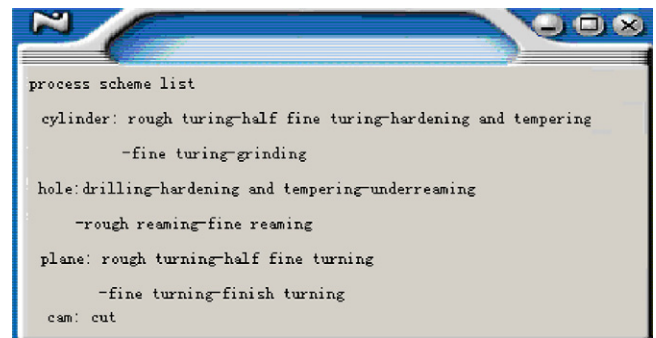


Fig. 13. Process scheme output interface.

kind of reasoning method, form features are stressed on. The inference procedure is shown in Fig. 8, where j = total number of form features.

4. System development

4.1. Knowledge-based approach

A knowledge-based approach has been used for developing DAPPS. Generally, structure manufacturability evaluation and process planning are mostly experience based, so it is significant to use the knowledge-based engineering (KBE) approach to aid the inexperienced process preparer and save

sufficient time for experienced preparer to consider creative technology. The knowledge-based approach involves knowledge acquisition, knowledge representation and construction of the knowledge bases and user interfaces. Analytical or empirical knowledge from 3D designers, process designers, handbooks, research journals, technology documents are the main sources of rules to build the knowledge base. The knowledge of DAPPS is divided into two levels. One is administrative knowledge and the other is technique knowledge. Both are derived from production rules based on which is easier and more popular to represent knowledge. The production rules are usually of the IF-THEN kind. For example, certain product structure manufacturability rules are as follows [13]:

Product-structure-manufacturability_evaluation_Rule_1
 IF Molding. Radius < (Molding. Thickness1 + Molding. Thickness2)/3.0
 THEN molding round angle is too small and easily cracks!

4.2. Prototype development

The system prototype has been developed using Visual Basic in a windows environment with Microsoft Access 2000 as a database support. Some prototype screen pictures are shown in Figs. 9–13.

5. Conclusions

The work reported here on DAPP is a part of a digitization system developed by the authors. This research seeks to establish systematic strategy and a system tool for the practice

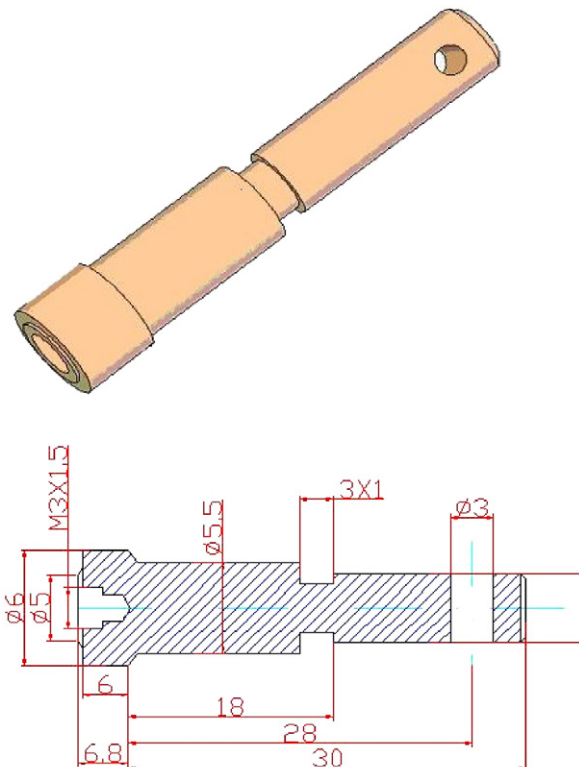


Fig. 11. A typical part TYY-02-05.

of digitization agile process preparation. An initial round of investigation has been completed, and a prototype has been developed. The next steps are to improve the function of process information abstraction with more automaticity and less information loss and put the prototype system into actual application in demonstration enterprise. Findings from the ongoing investigation will be reported separately in the near future.

Compared with literature survey, the main tasks of this research are as follows:

- (a) After analysis of the characteristics of the process preparation system (DAPPS), the needs for DAPPS are explored in depth.
- (b) A framework is proposed to support DAPPS by providing computer-aided and database environments for DAPPS, advisory tools for product structure manufacturability evaluation, rapid process planning, tooling design, process consumption determination as well as field process preparation, and knowledge bases incorporated in the advisory tools.
- (c) According to the framework, a design support strategy for DAPPS is investigated to guide the coordination of preparation activities.
- (d) Based on the proposed approach and strategy, the system prototype has been developed employing the knowledge-based approach.

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